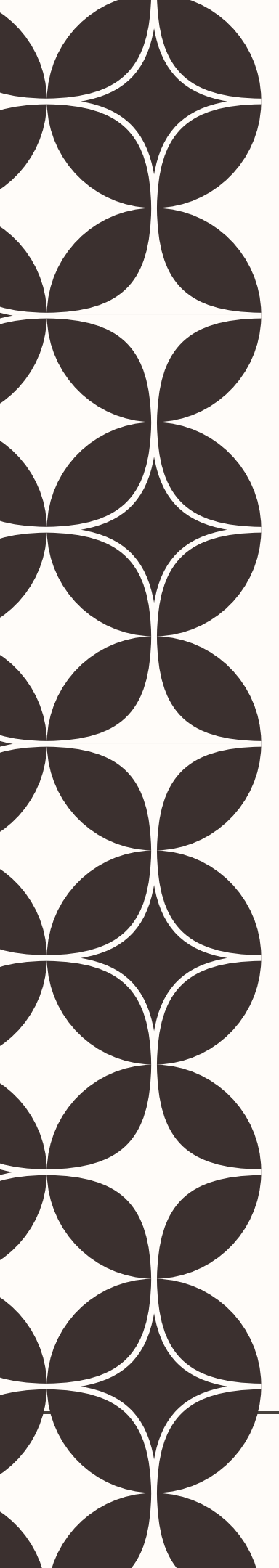


Predicting Youth METRO Ridership after the Youth Cruz Free Program

Intro to Statistical Learning – Final Presentation

By: Mila Lewis –Peinado

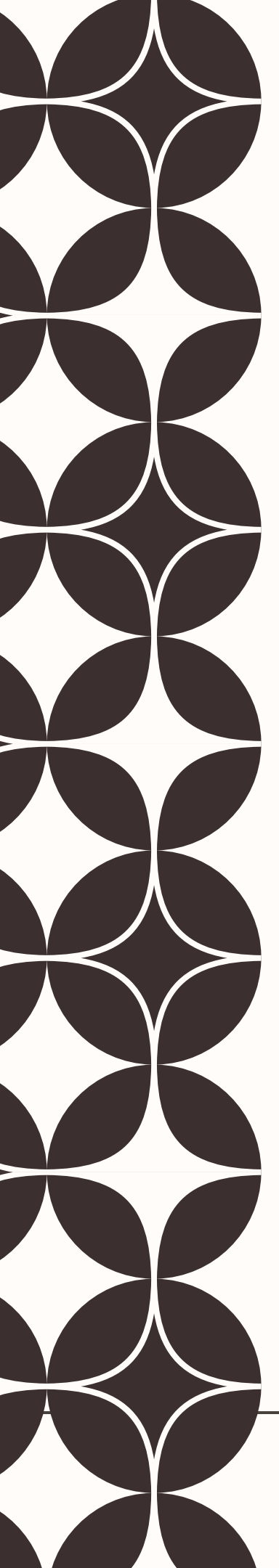




Introduction

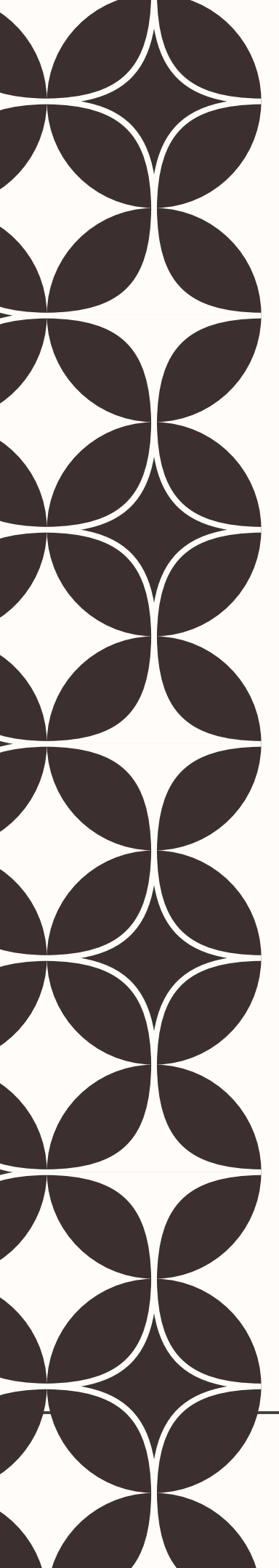
- Youth Cruz Free program by Santa Cruz METRO in 2023
- 2 rounds of surveys among local high schools
 - How program changed travel behavior; classification
 - 127 variables: sociodemographic status, lifestyle habits, perceptions of transit, etc..
 - 392 observations

Which factors best predict whether the Youth Cruz Free program increased a student's METRO usage?



Methods

- Reduced predictors to 11, removing NAs brought observations down to 152
 - focus on activity frequency
 - Lasso, Ridge, & Elastic Net Regression
 - goal: predictive modeling + feature selection
 - k-fold cross validation
 - class imbalance → decision threshold to 0.25
 - Ridge had AUC = 0.82
- 



Discussion

- Important predictors
 - Positive
 - already rode METRO during the program
 - work frequently
 - Negative
 - attend school fewer days per week
 - play school sports often
 - Future research: more survey data & casual inference
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